Predicting Student Academic Performance Using Machine Learning

Machine learning: Fundamentals and Applications (AAI-510-01)

Group 3 Final Project

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BUSINEES PROBLEM

Challenges
Schools often identify struggling students only after academic performance has already declined.

Consequences

- Missed intervention opportunities
- Lower student success rates
- Inefficient allocation of support resources
- Increased risk of long-term academic failure

Goal
Develop a machine learning model that predicts which students are academically at risk so educators can intervene earlier.

DATASET OVERVIEW

Student Performance Dataset

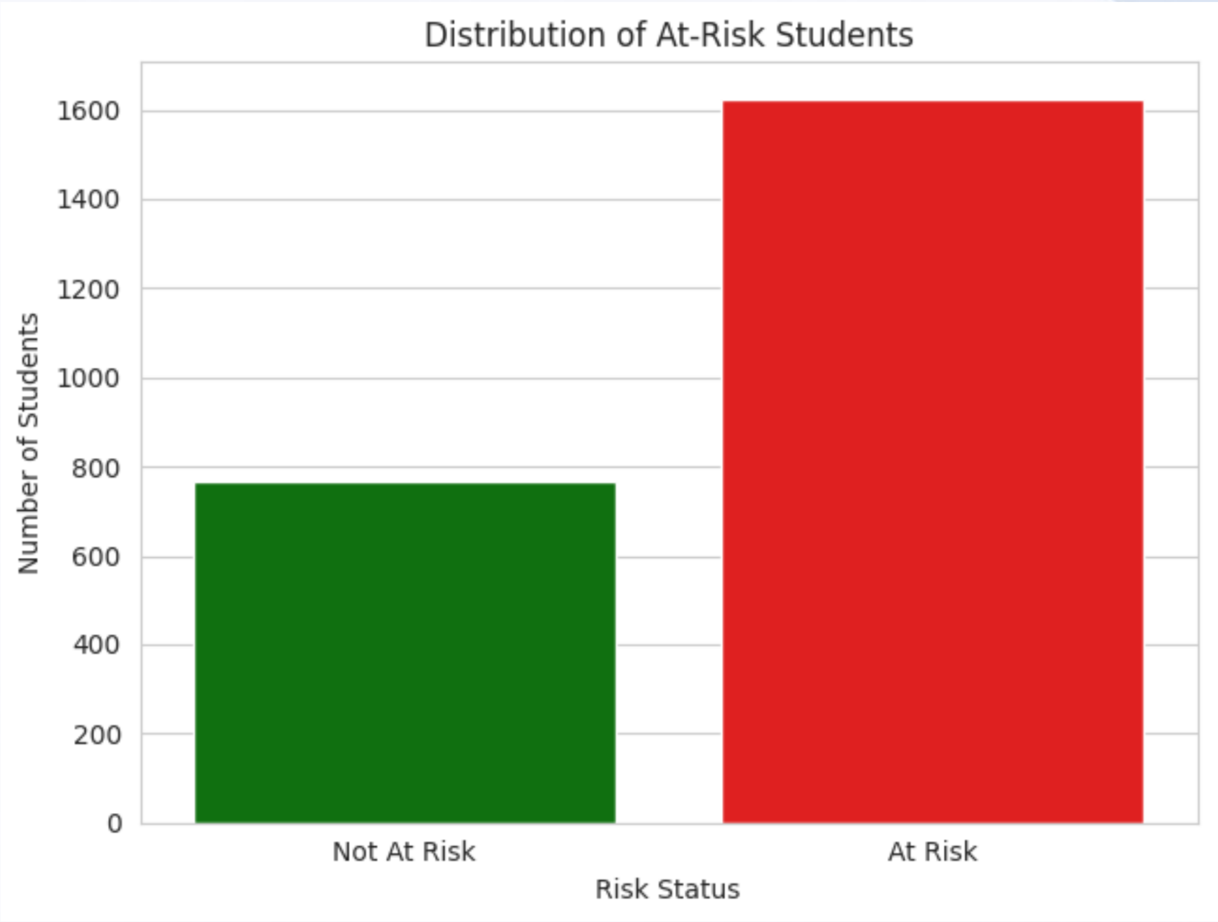
- ➢ 2,392 student records
- ➢ Academic and behavioral information
- ➢ Demographic and support-related factors

Examples of Features

- ❑ Attendance
- ❑ Study Time
- ❑ Tutoring
- ❑ Parental Support
- ❑ Extracurricular Activities

Business Outcome

Predict whether a student is academically at risk.



source: Notebook-Step7.1

DATA UNDERSTANDING & KEY INSIGHTS

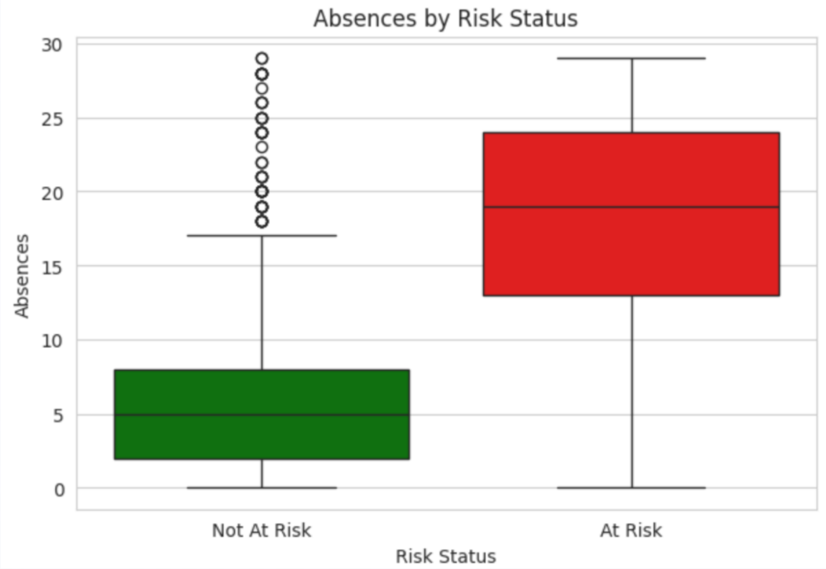
Key Findings From Exploratory Analysis

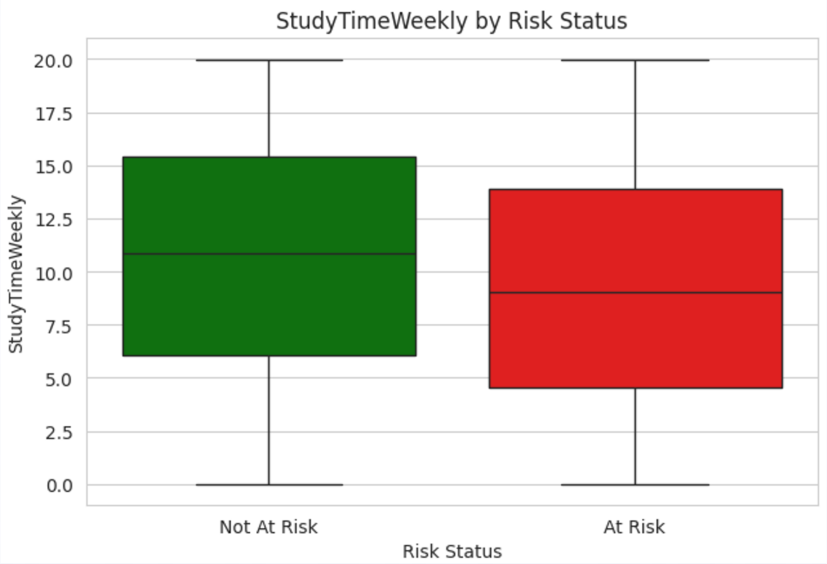
Students At Risk Tend To

- ❑ Miss more classes
- ❑ Study fewer hours
- ❑ Receive less parental support
- ❑ Participate in fewer academic support activities

Most Important Observation

Attendance and study behavior consistently showed strong relationships with academic performance.





DATA PREPARATION & FEATURE SELECTION

Preparing Data for Machine Learning

Data Quality

- ➢ No significant missing values
- ➢ No duplicate records
- ➢ Clean dataset structure

Removed Features

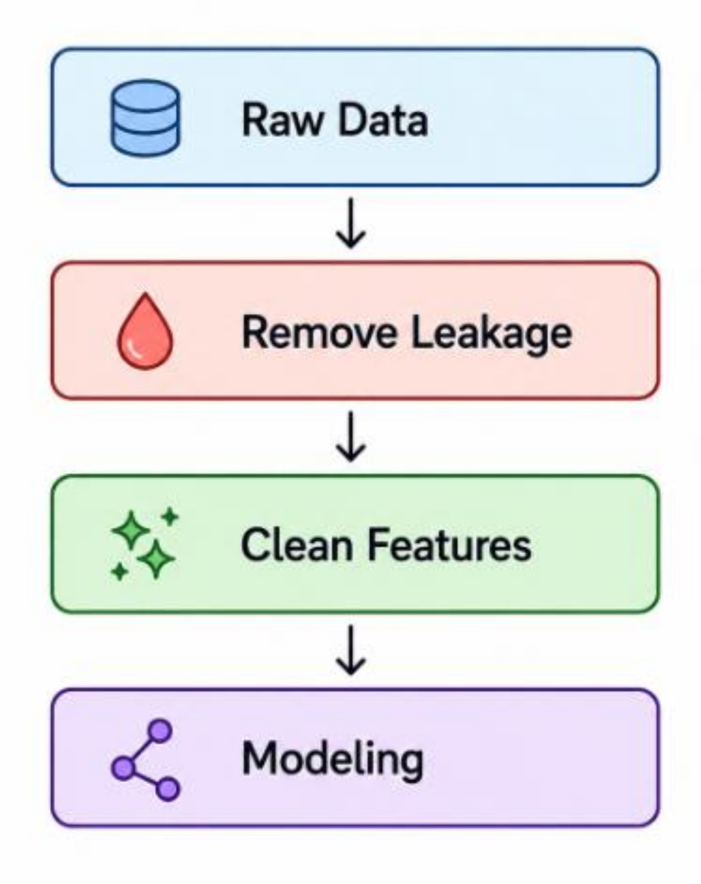
- ❑ StudentID
- ❑ GPA
- ❑ GradeClass
- ❑ GradeClassLabel

Why?

Prevent data leakage and ensure realistic predictions.

Final Dataset

Focused on information available before final academic outcomes occur.



MACHINE LEARNING MODELS EVALUATED

Models Evaluated

Model	Purpose
Decision Tree	Baseline model
Random Forest	Ensemble model
XGBoost	Advanced boosting model

Evaluation Approach

- 80/20 train-test split
- 5-fold stratified cross-validation
- Multiple performance metrics

MODEL PERFORMANCE COMPARISON

Which Model Performed Best?

Results Summary

	Model	CV Accuracy	CV Recall	CV F1-Score	CV ROC-AUC	Test Accuracy	Test Precision	Test Recall	Test F1-Score	Test ROC-AUC
0	Decision Tree	0.8657	0.9338	0.9043	0.8920	0.8518	0.8735	0.9138	0.8932	0.8826
1	Random Forest	0.8887	0.9392	0.9198	0.9225	0.8768	0.8844	0.9415	0.9121	0.9235
2	XGBoost	0.8907	0.9423	0.9214	0.9239	0.8894	0.8953	0.9477	0.9208	0.9294

Findings

- All models performed well
- Ensemble methods outperformed the Decision Tree
- XGBoost achieved the strongest overall performance

XGBOOST OPTIMIZATION

Improving the Best Model

Optimization Process

- Hyperparameter tuning
- Cross-validation
- Performance validation

Result

Optimized XGBoost improved predictive reliability and became the final recommended model.

FINAL MODEL PERFORMANCE

Improving the Best Model

Final XGBoost Results

```
Optimized XGBoost Test Performance
--------------------------------------------------
Accuracy : 0.8894
Precision: 0.8931
Recall    : 0.9508
F1-Score : 0.9210
ROC-AUC   : 0.9311
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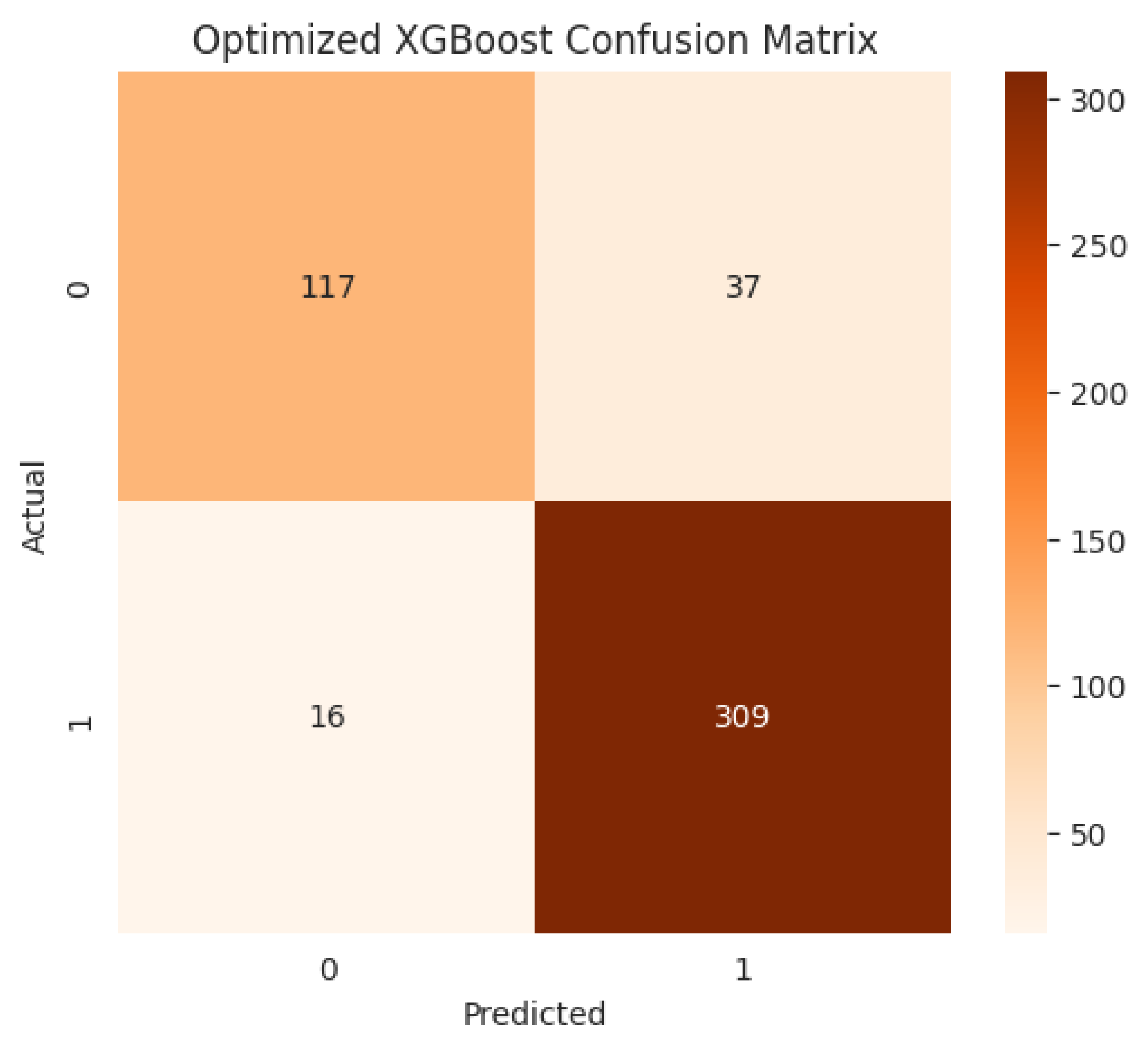
Key Business Metric

95% Recall

The model successfully identifies most at-risk students.



WHAT DRIVES STUDENT RISK?

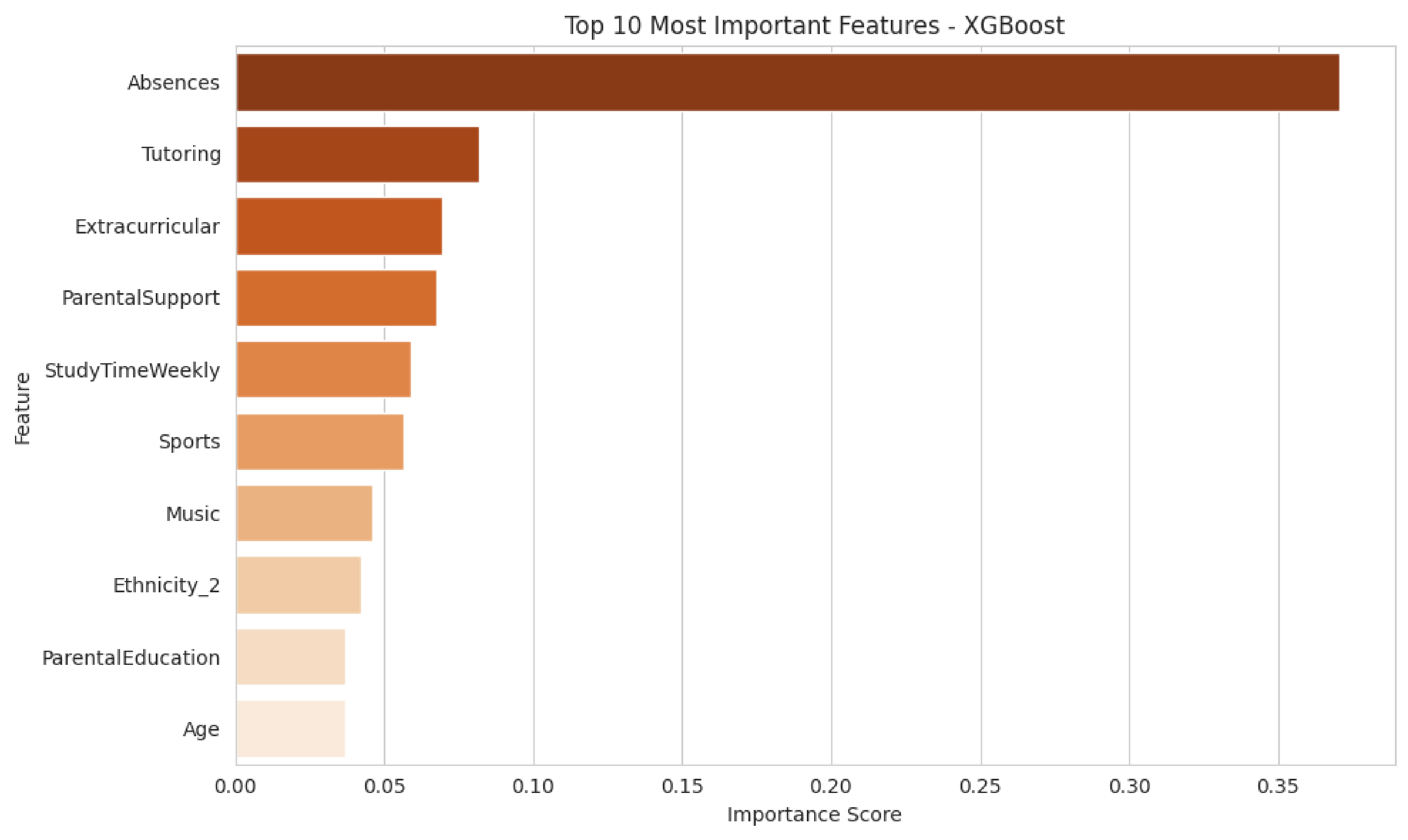
Most Important Risk Factors

Top Predictors

- Absences
- Study Time
- Parental Support
- Tutoring Participation
- Educational Engagement

Business Insight

Student success is strongly influenced by attendance, study habits, and support systems.



RISKS, ETHICS & DEPLOYMENT

Risks and Deployment Strategy

Risks

- Potential prediction bias
- Data privacy concerns
- Overreliance on automated decisions

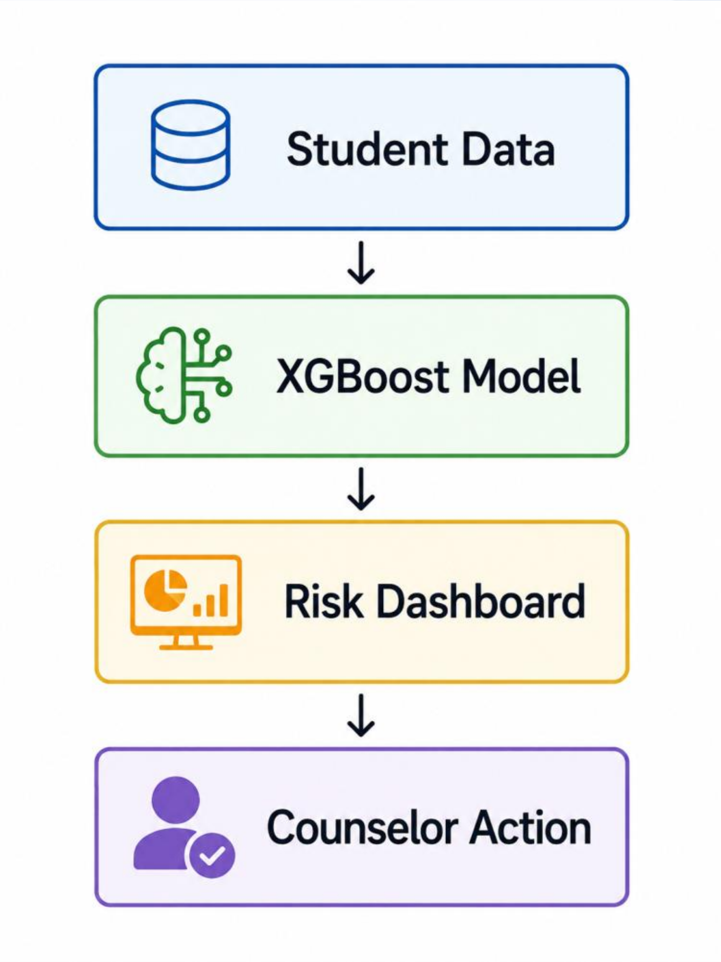
Mitigations

- Human review of predictions
- Regular model monitoring
- Student data protection policies

Deployment

Weekly batch prediction system

Student Data → XGBoost → Risk Dashboard → Counselor Intervention



Recommendation & Conclusion

Final Recommendation
Deploy the Optimized XGBoost model as an early-warning system for identifying academically at-risk students.

Benefits

- ❑ Earlier intervention
- ❑ Better resource allocation
- ❑ Improved student outcomes
- ❑ Data-driven decision making

Final Result
95% Recall
Strong ability to identify students needing support.